SE LAB 3

COMPONENT MODELLING & ARCHITECTURAL PATTERN SELECTION

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Architecture Selection:

We chose Microservices Architecture for the Centralized Audit Trail Compliance Engine.

Architectural choice

The system is organised into independent services: an Audit Log Ingestion Service, PII Masking Service, Audit Storage Service, Integrity & Hashing Service, Compliance Query Service, and Immutable Audit Export Service. Each service is responsible for a specific function and communicates with other services through APIs, allowing the ingestion-to-masking-to-storage-to-verification workflow to be clearly separated.

Two scenario-related reasons

1. **Independent scalability:** The system must support complex filtering across 10 million historical audit records in under 1 second. Separating the Compliance Query Service from the other services allows query processing to be scaled independently when the number of audit queries increases, without scaling the entire compliance engine.
2. **Fault isolation and independent deployment:** The system performs different security-critical operations such as log ingestion, PII masking, integrity verification, storage, querying, and audit exporting. Separating these functions into independent services means that a failure or update in one service does not require the entire system to be stopped or redeployed. This also makes individual services easier to test and maintain.

Security advantage

Because PII masking is handled by a dedicated PII Masking Service before audit records are persisted, sensitive information such as Social Security Numbers and credit card numbers can be consistently removed or masked before reaching the storage layer. The separate Integrity & Hashing Service can also generate SHA-256 integrity hashes, helping verify that stored audit records have not been modified.

Performance benefit

Since the Compliance Query Service can be independently scaled and optimized for complex filtering, it can handle large volumes of historical audit records without affecting log ingestion, PII masking, or export operations. This helps the system meet the requirement of querying up to 10 million historical records in under 1 second.

UML Component Diagram:

Centralized Audit Trail Compliance:

